Simulation-based inference with deep learning suggests speed climbers combine innovation and copying to improve performance

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Abstract

In the Olympic sport of speed climbing, athletes compete to reach the top of a 15-meter wall as quickly as possible. Since the standardization of the speed climbing route in 2007, improvement has been driven by a process of cumulative cultural evolution—new route sequences innovated by some are copied and improved upon by others. In this study, we use simulation-based inference to fit an agent-based model of speed climbing to 12 years of competition times (2007-2019). In the fitted model, innovation and copying are used roughly equally by climbers, with copying having only a slight advantage. Slower agents are more likely to innovate, likely in pursuit of strategies that give them a competitive advantage. Population size negatively predicts innovation, presumably because innovation is not as useful when there are plenty of existing solutions to choose from. An additional analysis of real route sequences from 2012-2019 World Championships suggests that other factors, like height and weight, influence the use of some holds, but may not affect performance or the probability of adopting innovations. Finally, the model suggests that climbers may not be using innovation and copying optimally—continued improvement may require more emphasis on innovation and less reliance on copying.

Keywords: speed climbing, cumulative cultural evolution, social learning strategies, innovation, sports, simulation-based inference

Introduction

The 2020 Tokyo Olympics games put speed climbing on the world stage for the first time. On a 15-meter wall with 20 standardized holds, the goal is simple: get to the top as fast as you can (Figure 1A). Speed climbing is one part of a three-stage competition at the Olympics, which also includes bouldering and lead climbing. Where bouldering and lead climbing are scored on how far climbers advance along a route, speed climbing is scored on time alone. When the speed climbing route was standardized in 2007, climbers used every hold and had a more-or-less blank slate of route variations to explore. By 2019, the final year of continuous competition before the COVID-19 pandemic, there was a wide range of route variations (1) that skipped up to a quarter of the handholds on the wall (2), along with a dramatic decrease in the average time spent on each hold. In the intervening years, the fastest climbing times have improved by 15.9 seconds for women (2007: 23 seconds; 2019: 7.1 seconds) and by 9.9 seconds for men (2007: 15.4 seconds; 2019: 5.5 seconds) (Figure 1B).

Speed climbing athletes have two options: they can copy the route of the fastest competitors in previous competitions, aiming to outcompete the leaders through athletic ability, or they can search for an undiscovered solution to send them straight to the top. This choice between copying and innovating among athletes makes speed climbing a fascinating example of cumulative cultural evolution—iterative improvement in performance through innovation and copying (3). In this paper, we define innovation as the discovery of new solutions via individual learning (i.e., exploration), and copying as the adoption of existing solutions via social learning (i.e., exploitation). A famous example of innovation occurred when Reza Alipour skipped the fourth handhold on his way to breaking the speed climbing world record in 2017 (4,5). This route became known as "the Reza" and spread rapidly through the competitive climbing community[1]. Just two years later, Tomoa Narasaki introduced a variation on "the Reza" that skips an additional hold (4,5). The "Tomoa skip" is now the dominant opening sequence, with 92% of athletes using it in the 2020 Olympic finals (2). The history of speed climbing is one of cumulative improvement through "leaps" and "tweaks" (6)—major innovations (i.e., "leaps", pun intended) are rapidly adopted and improved upon (i.e., "tweaks") over time.

[1] https://www.redbull.com/int-en/episodes/up-to-speed-reel-rock-s05-e01

Humans' ecological success as a species is heavily driven by our ability to flexibly innovate and copy to solve problems (7). Theory predicts that a combination of sporadic innovation and widespread copying is often adaptive for groups because it allows new solutions to be rapidly adopted without compromising stability (8). However, group incentives often conflict with individual incentives. Simulations show that competition increases pressure to discover unexplored solutions (9), and real-world studies in competitive contexts like computer-based tournaments (10,11), board games (12), and sports coaching (13) demonstrate that people often use a more balanced combination of personal and social information. This conflict likely occurs in speed climbing, where copying the fastest route will improve the average time of all competitors, but a successful innovation tends to make an individual the fastest.

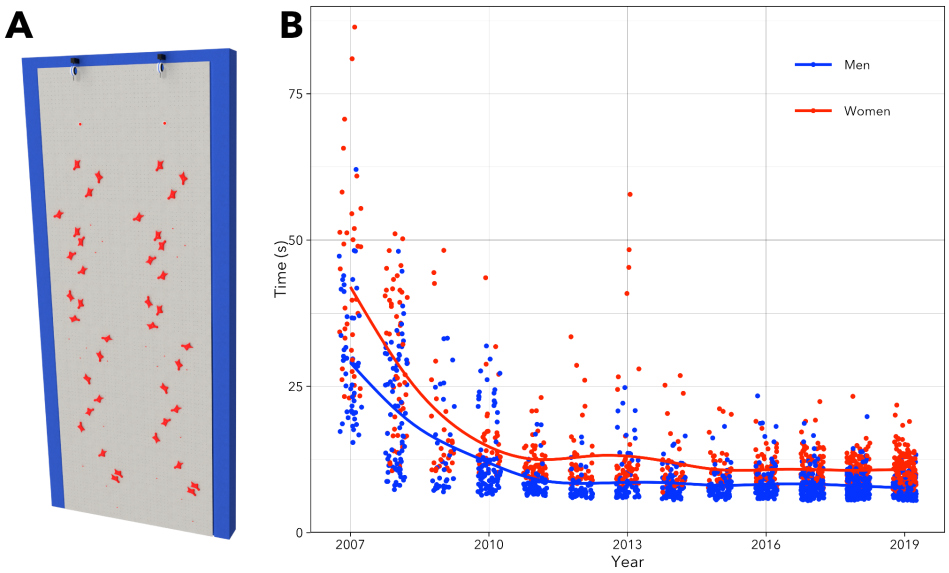


Figure 1. A) The standardized climbing wall for speed climbing as of 2007. Side-by-side walls allow for two races to occur at once. B) The best time at each world competition between 2007 and 2019.

People also show a great deal of flexibility in their social learning strategies (14). In programming tournaments where people can access and improve upon previous solutions, participants exhibit a wide range of behavior—from pure innovation and pure copying to a spectrum of intermediate combinations (15). Similarly, in the opening moves of the board game Go, players have been shown to use a combination of personal information and social information (12), with comparable patterns seen in football formations (13). Some strategies, like prestige or success bias (i.e., copying those who have won previously), are likely to be widespread across competitive contexts (12,15–17). Other strategies, like choosing whether to innovate or copy based on one's current performance, might be more variable. Theoretically, an "innovate-when-underperforming" strategy should be effective under a lot of conditions (18–21), and high performers appear to exhibit more or less copying depending on the context (12,15). Climbers may also adjust their strategies based on their ability level, as has been seen in board games and programming contests—a faster climber may be more likely to stick with their current solution, while a slower climber may be more likely to search for a better one.

Another factor that influences cumulative improvement is population size (22). Research has shown that population size can have far-reaching effects on cultural variation and complexity (23,24). Islands with larger populations have more tool complexity (25), and larger cities tend to have more patents (26). Put simply, growing populations have an increasing capacity to discover and carry new solutions. Speed climbing has expanded from 91 competitors in 2007 (38 women, 53 men) to 314 in 2019 (208 women, 171 men) (Figure 1B), while climbing times have dropped and route diversity has increased (1). Such dramatic changes in population size and solution diversity have probably changed the payoffs for innovation and copying. Today, there are more route variations produced by more people, which could boost the benefits of copying relative to innovating. In other contexts, innovation rate appears to be boosted by population size (24), people are more likely to copy when there are more solutions in the population (11), and larger populations get more of a benefit from recombining existing solutions (27).

The aim of this study is to describe the innovation and copying dynamics that drive cumulative improvement in competitive speed climbing. To do this, we use simulation-based inference to fit an agent-based model to 12 years of competitive speed climbing times. Simulation-based inference (a.k.a., generative inference) is a method for recovering the parameter values of simulations that most closely recreate real data—allowing one to make inferences about the processes generating patterns without direct observation. We use it to determine the precise combination of innovation, copying, and the strategies governing the use of innovation and copying, that may have led to the temporal distribution of climbing times in the historical record. In recent years,

simulation-based inference has been used to characterize cultural dynamics in a variety of contexts including music (28), birdsong (29–31), and social media (32,33). Ideally, one would fit an agent-based model to both climbing times and route sequences. Researchers have used emerging technologies like motion tracking to "transcribe" route sequences from a handful of climbing competitions (4,5,34–37), but, to our knowledge, a large-scale, longitudinal dataset of route sequences in speed climbing does not yet exist. Here, we fit the agent-based model to climbing times alone, and use a descriptive analysis of a much smaller set of manually-transcribed route sequences from IFSC World Championships to validate the output (Figure 2).

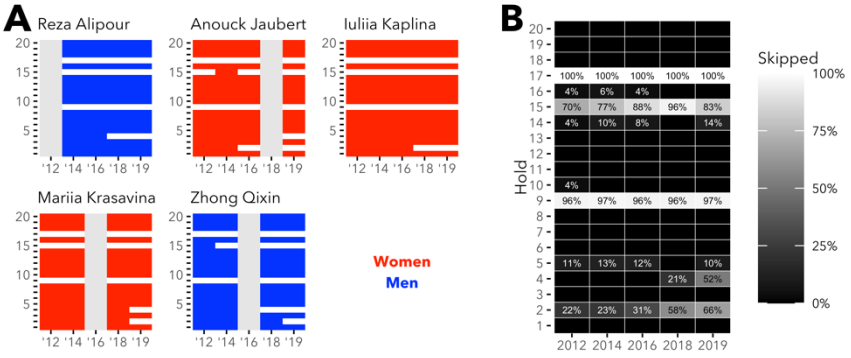


Figure 2. A) Route sequences used by five high profile competitors in the five IFSC World Championships between 2012-2019, manually-transcribed for this study (see Methods). White horizontal bars denote holds that are skipped, and large grey vertical bars denote years in which they did not compete. B) A heatmap of the most commonly skipped holds among all participants in the same competitions.

Methods

Data

For simulation-based inference, we analyzed the speed climbing results from all International Federation of Sport Climbing (IFSC) World Cups and World Championships that occurred between 2007 and 2019, totalling 87 different competitions with 7,724 speed climbing runs from 971 athletes (564 men, 407 women). The majority of these data are from a pre-existing GitHub repository (https://github.com/lfigil/ifsc-data) that scraped results from the IFSC website (https://ifsc.results.info/) on February 1, 2024. We checked the coverage of this repository and found that it included all events except for the World Championship in Aviles, Spain in 2007, which we manually added to the data ourselves. The results of the World Championship held in Qinghai in 2009 were incorrectly logged by the IFSC (ranks instead of times: https://ifsc.results.info/#/event/568/general/speed), so the entire event was dropped from the data. Besides that, there was only one impossible climbing time—an exactly 2 second run in Poland in 2009, also likely a rank instead of time—that was dropped from the data.

To validate the findings and assumptions of our model, we also manually transcribed route sequences from the finals of the IFSC World Championships in 2012, 2014, 2016, 2018, and 2019—all years in our study period with public video documentation of all competitors. In the vast majority of cases, only the first round of finals was included, to maximize sample size (all 32 participants across men and women) without introducing effort as a confound. In a single "wildcard" case, where the athlete's opponent was disqualified and they did not have to complete the route, we transcribed their sequence from the second round of finals. Falls were excluded, resulting in 137 speed climbing runs from 67 athletes (32 men, 35 women) across the five competitions. For each athlete, we also collected their age, career start date, height (cm), and weight (kg)—when available—from public sources like Wikipedia, Digital Rock, and the IFSC website.

Agent-based model

The agent-based model simulates a population of professional speed climbers who improve in their athletic performance, innovate new route sequences, and copy others' routes sequences according to a set of rules. Women and men copy route sequences from one another (2), so all athletes were included in a single model to avoid confounding between-gender copying with within-gender innovation.

Each agent in the model enters and exits the sport in the same years as a real climber, so that the dynamics of population growth and turnover exactly match those in the real dataset between 2007 and 2019. Agents are initialized with three things that determine their record time in a given year: (1) a reference time, or the baseline amount of time they spend on each hand hold, (2) a numeric vector that controls their rate of athletic improvement over time, and (3) a table that stores their initial route sequence. The reference times come from real climbing times in 2007 divided by 20: the number of hand holds in the standardized route. The level of athletic improvement in a given year (δ_y) comes from a bounded exponential decay function:

$$\delta_y = (1 - \phi)\left(\frac{\alpha}{\alpha^{y-2007}}\right) + \phi$$

where y is the year, a is the rate of improvement and φ is the asymptotic lower bound. Exponential decay functions are commonly used to study improvement in sports, which generally plateaus over time (38,39). a and φ vary for women and men to allow for differences in athletic improvement. a_w, a_m, φ_w, and φ_m are parameters that are estimated during simulation-based inference.

Route sequences in the model are composed of a binary vector (ϱ_h) that stores whether or not each hold (h) is used (e.g., ϱ_h = {0, 1, 1… 0}), and a numeric vector ($\varkappa_h$) that controls the amount of time spent on each hold (e.g., $\varkappa_h$ = {1.15, 0.72, 0.96… 1.03}). We assume that agents use all holds in the first year (ϱ_h = {1, 1, 1… 1}). $\varkappa_h$ is drawn from a truncated normal distribution with a lower bound of 0, a mean of 1, and a standard deviation of 0.5, which generates a distribution of times per hold that is similar to those observed in both speed climbing (36) and lead climbing (37). To compute the record time of an agent (i) in a particular year (y), we multiply their reference time (r_i) by their level of athletic improvement (δ_y) and the sum of the time multipliers ($\varkappa_h$) from the holds that they use:

$$t_{yi} = r_i \delta_y \sum_{h=1}^{20} \rho_{hi} \kappa_{hi}$$

At the beginning of each year in the model, each agent in the population can innovate with probability μ or copy the preferred route of another agent with probability λ. Innovations are changes to the binary vector ϱ_h where a 1 is flipped to a 0. The first hold cannot be skipped because climbers are required to start on it. The only other hard constraint on innovation is that agents cannot traverse more than 1.65 meters—the distance of the "Reza" (i.e., skipping hold #4)—without touching a climbing hold.

Copying occurs in two steps. First, agents compile the preferred routes of a proportion of the best climbers in that year (τ). Then, they compute the overall times (t_{yi}) from each route, including their own, and keep the sequence (ϱ_h) and time multiples ($\varkappa_h$) from the best option among those routes. Note that copying occurs before innovation, so agents can copy a new route and then modify it in a single timestep.

We are also interested in potential interactions between innovation and copying probability and other factors. To model these, we use logit transformations to compute individual probabilities of innovation and copying (μ_i and λ_i) that are modulated by the year (y), population size (n_y), and the agents' current record time (t_{yi}).

$$logit(\mu_i) = logit(\mu) + \beta_{\mu t} t_{yi} + \beta_{\mu n} n_y + \beta_{\mu y}(y - 2007)$$
$$logit(\lambda_i) = logit(\lambda) + \beta_{\lambda t} t_{yi} + \beta_{\lambda n} n_y + \beta_{\lambda y}(y - 2007)$$

y, n_y, and t_{yi} are all z-scaled prior to transformation so that the coefficients (β) are comparable in the fitted models. y and n_y were both included because population size has increased over time—we cannot estimate the effect of the latter without controlling for the former. $\beta_{\mu y}$ and $\beta_{\lambda y}$ introduce time trends in innovation and copying (i.e., increases or decreases in innovation and copying over time), $\beta_{\mu n}$ and $\beta_{\lambda n}$ introduce interactions between innovation and copying and population size, and $\beta_{\mu t}$ and $\beta_{\lambda t}$ introduce interactions between innovation and copying and agents' current performance. Note that the effect of population size on innovation and copying here is in addition to the indirect effect of it on the spread of existing route sequences, which occurs organically as the model recapitulates the entry and exit of real climbers into and out of the sport.

The output of the agent-based model is the record time of each agent in each year from 2008 to 2019.

Simulation-based inference

The prior distributions for each parameter can be seen in Table 1. Simulated distributions of climbing times from these prior distributions can be seen in Figure 3A.

Parameter	Prior	Lower Bound
μ: innovation probability	uniform(0, 1)	NA
$\beta_{\mu y}$: time trend	normal(0, 1)	NA
$\beta_{\mu l}$: innovation x current record	normal(0, 1)	NA
$\beta_{\mu n}$: innovation x population size	normal(0, 1)	NA
λ: copying probability	uniform(0, 1)	NA
$\beta_{\lambda y}$: time trend	normal(0, 1)	NA
$\beta_{\lambda l}$: copying x current record	normal(0, 1)	NA
$\beta_{\lambda n}$: copying x population size	normal(0, 1)	NA
τ: proportion of top climbers	uniform(0, 1)	NA
a_w: rate of improvement (W)	normal(2, 0.5)	1
a_m: rate of improvement (M)	normal(2, 0.5)	1
φ_w: lower bound of improvement (W)	uniform(0, 1)	NA
φ_m: lower bound of improvement (M)	uniform(0, 1)	NA

Table 1. The prior distributions for each parameter in the agent-based model. The parameters in italics are time trends and interactions between innovation and copying and the current records and population size, respectively.

Simulation-based inference was conducted using BayesFlow (40), a method that uses amortized neural networks to approximate the posterior distributions of parameters in a generative model (41). We used BayesFlow because it has been shown to outcompete various forms of approximate Bayesian computation (e.g., with MCMC, random forests, neural networks) in both accuracy and computational cost (41,42), and has already been used to infer social learning strategies (30). Simulation-based inference was conducted in the following steps:

1. 100,000 iterations of the agent-based model were run with parameter values sampled from the prior distributions in Table 1. The output of the agent-based model is a distribution of record times for each year from 2007 to 2019. The results were split into a training set (n = 99,000) and a testing set (n = 1,000).
2. BayesFlow was run for 200 epochs with a batch size of 32 and a learning rate of 0.0005. Informative summary statistics were learned from the raw model output using a time-series transformer (43) with four hidden layers of 512 units using a rectified linear unit activation function (ReLU). These summary statistics were then mapped to the parameters of the agent-based model using an invertible neural network with 10 affine coupling layers, each of which has four hidden layers of 128 units using ReLU with a dropout probability of 0.05. This architecture is based on default modules that come with BayesFlow.
3. The trained neural networks were used to produce 10,000 posterior predictions for each parameter in the agent-based model, based on the annual distribution of record times in the real data.

Diagnostic plots can be seen in Figures S1 to S3. We found that the inference procedure can recover the known values of innovation probability, copying probability, and the lower bound of athletic improvement for men and women from the testing set with high accuracy (R^2 = 0.70, 0.76, 0.92, and 0.91, respectively; Figure S2). The rate of athletic improvement for men and women is recoverable with fairly good accuracy (R^2 = 0.58 and 0.54). The interactions between innovation/copying and time, current ability, and population size are also recoverable, although to a lesser extent (R^2 ~ 0.21-0.61). The number of top climbers that athletes copy from is not recoverable from the data (R^2 = 0.02), but we kept it as a varying parameter in the model to avoid biasing our estimate of copying probability. The loss curve steadily decreases over time and stabilizes before the end of training, indicating that BayesFlow has effectively learned to estimate parameter values from the agent-based model (Figure S3). We initially explored two other parameters that introduced biases in innovation (ϵ_D and ϵ_R, see Agent-Based Model in Methods), but chose to exclude them because they were not recoverable by BayesFlow and reduced our ability to infer the other parameters (Figure S1).

Manually-transcribed route sequences

In addition to the main analysis, we conducted three statistical analyses of the manually-transcribed route sequences from IFSC World Championships. The goal of these analyses was to assess whether demographic factors that were not included in our agent-based model play a major role in overall performance, hold usage, and copying—specifically the adoption of the "Reza" (skipping hold #4), the only innovation that appears in our sequence data (see Figure 2).

1. First, we ran a linear model with climbing time as the outcome variable, with age, career length, height, and weight as predictor variables, and with name, gender, and year as varying intercepts.

2. Second, we ran a multivariate logistic model with whether each hold was used as a set of binary outcome variables, with age, career length, height, weight, and gender as predictor variables, and name and year as varying intercepts. Only the eight holds with variation in use were included (#2, 4, 5, 9, 10, 14, 15, and 16). No significant multicollinearity was detected (variance inflation factors < 2).
3. Third, we ran a logistic model with whether hold #4 was used in 2018 and/or 2019 as the binary outcome variable (0 = the "Reza"), with age, career length, height, weight, and rank in 2016 as predictor variables. For this model, we used a subset of the manually-transcribed data. We only considered athletes who were present both in 2016 and in 2018 and/or 2019, excluding Reza Alipour, who introduced his signature move in 2017.

All three models were Bayesian multilevel models run in Stan using the brms package in R (44), fit using Markov chain Monte Carlo (MCMC) run across eight chains of 5000 iterations each. We used default priors for all model parameters besides the main effects of predictors, which were assigned Gaussian distributions centered around zero with standard deviations of 2 and 1 for the linear and logistic models, respectively.

There were several variables with incomplete data: age (99% coverage), career start date (96%), height (66%), and weight (42%). We handled missing data using the multiple imputation approach recommended by Bürkner (44). Multiple imputation outperforms both single imputation (e.g., replacing NA with averages) (45) and complete case analysis (e.g., dropping incomplete rows from the data) (46,47), neither of which would be appropriate as data are not missing at random—higher performing climbers are more likely to have public data. First, we used random forest machine learning to generate 10 different imputed versions of our data using the mice package in R (48). Then, we fit our models to each imputed dataset separately and pooled the results across all of the models by simply combining the posterior distributions for each parameter (44). Model diagnostics for each of the three models applied to each of the 10 imputed datasets can be seen in Tables S1, S2, and S3.

Results

The posterior distributions for innovation and copying probability can be seen in Figure 2A and 2B. Copying appears to be slightly more likely than innovation in speed climbing, although both occur at a fairly high rate (copying: 0.37 [95% CI: 0.27-0.48]; innovation: 0.26 [95% CI: 0.17-0.36]). There are two reasons why this finding should be interpreted with caution. First, the estimated innovation and copying probabilities in the model are probabilities of attempts. Many attempted cases of copying in the later years will fail because agents have already converged to common sequences (i.e., none of the tested sequences will beat the current sequence), and cases of innovation will fail because there are no more options within the physical constraints. This is evident from the fact that the gap between *actual* innovation and copying probabilities (innovation: 0.24; copying: 0.29; averaged across 10,000 posterior simulations), calculated by dividing the total number of successful innovations and copying events by the total population size, is smaller than the *attempted* innovation and copying probabilities. Third, the model appears to be underestimating the true innovation rate, based on the discrepancy between the estimated and the actual number of holds skipped in 2019 (Figure 3B). That being said, the observed number of holds skipped in 2019 may be an overestimate, as the convenience sample comes from the inaugural Olympic speed climbing qualifiers.

The estimated interactions between innovation and copying probability and other factors can be seen in Figure 2A and 2B. After accounting for a negative time trend, innovation probability appears to be affected by both population size and an agent's current record time. Slower agents with high record times are more likely to innovate, but innovation becomes less likely as more athletes enter the sport. Copying probability, on the other hand, shows only a very weak time trend and seems to be unaffected by population size and an agent's current record time.

The posterior estimates for the lower bound of athletic improvement are 0.268 (95% CI: 0.251-0.286) for women and 0.490 (95% CI: 0.469-0.514) for men, and for the rate of athletic improvement are 1.792 (95% CI: 1.733-1.855) for women and 2.574 (95% CI: 2.430-2.724) for men. This suggests that women have improved more than men—spending ~83% less time on hand holds in 2019 than in 2007, compared to ~51% for men—but at a slower rate.

Posterior simulations show that the fitted parameter values generate output that approximates the observed distribution of climbing times in 2019 (Figure 3A). The model slightly underestimates the number of holds skipped in 2019 (estimated median of four), although the observed value for both women and men (dashed line) may be an overestimate as it comes from the 2020 Olympics rather than a normal competition (Figure 3B). The most common route sequences from the posterior simulations can be seen in Figure 3C. The simulated sequences include commonly skipped holds among real climbers, such as the ninth and seventeenth holds, the upper starting hold (#2) and the "Reza" (#4)[2] (2).

[2] https://www.wired.com/story/the-best-speed-climbers-dash-up-walls-with-a-time-saving-move/

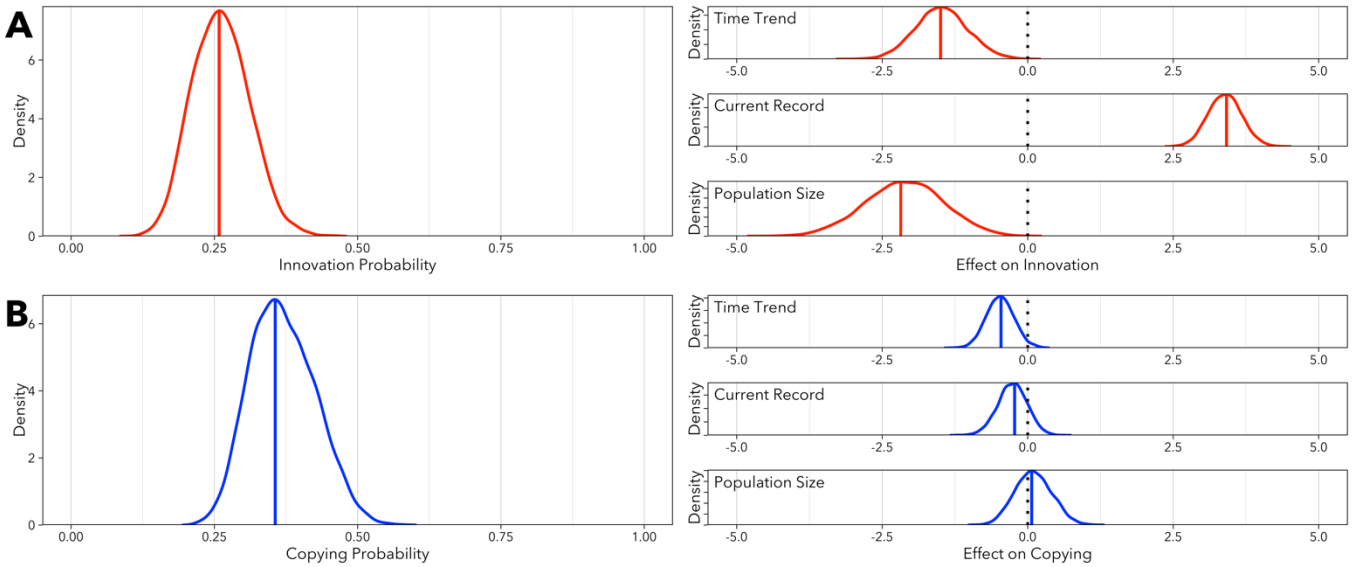


Figure 2. The posterior distributions for innovation probability (A: left), copying probability (B: left), and the time trends and interactions with population size and the current rank of climbers (A and B: right). All *y*-axes in the right panel have been normalized for visualization purposes.

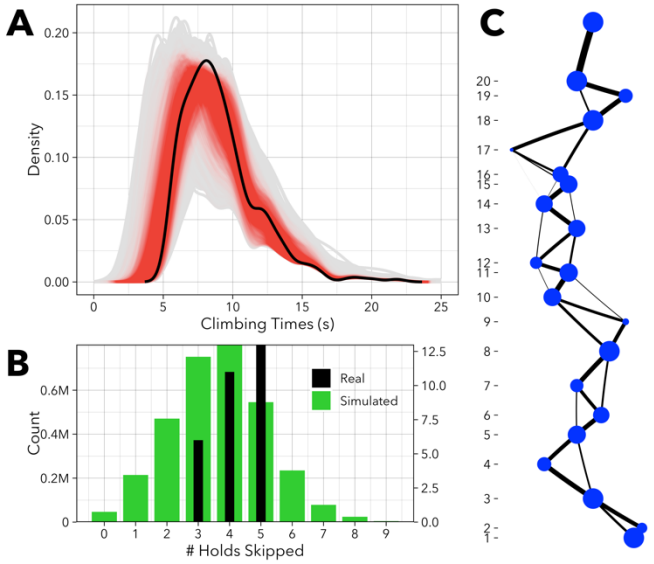


Figure 3. Posterior simulations from the model. Panel A shows the distribution of climbing times in 2019 from 10,000 prior simulations (red and blue), 10,000 posterior simulations (gray), and the observed data (black). Panel B shows the distribution of skipped holds in all route sequences in 2019 from 10,000 prior simulations in green, with the real distribution of skipped holds in the 2019 IFSC World Championship in black. Panel C shows all route sequences in 2019 from 10,000 posterior simulations. The thickness of the black line corresponds to transition probability, the size of the nodes corresponds to the probability that the hold is used (larger node is higher probability).

Figure 4 shows the results of posterior simulations from the fitted models across various levels of innovation and copying. From left to right, the columns show how different levels of innovation and copying lead to different minimum times, different average times, and different numbers of unique routes in 2019. In other words, the redder regions in the left column are optimal for the very best agents, the redder regions in the center column are optimal for average agents, and the bluer regions in the right column are optimal in terms of the overall diversity of route sequences. The black dots mark the estimated level of innovation and copying from the fitted model.

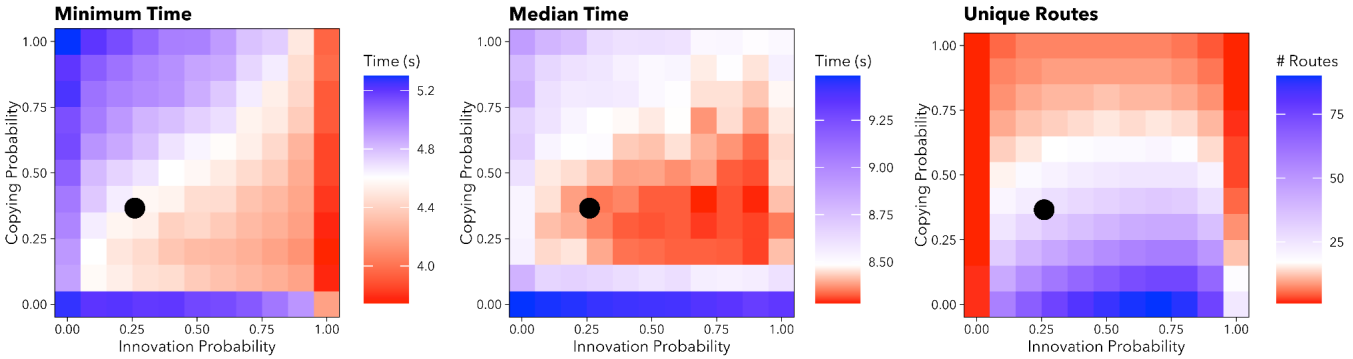


Figure 4. Posterior simulations from the fitted models assuming 11 different levels of innovation probability and copying probability. From left to right, the color of each cell in each column shows the average minimum time, the average median time, and the number of unique routes in 2019, calculated from 1,000 simulations for each combination of innovation and copying. The black dots mark the estimated combination of innovation and copying from the fitted model. The middle point for each heatmap (white) is set to the median value of the whole sample.

The optimal zones for the very best agents (left column of Figure 4) and average agents (middle column of Figure 4) have a great deal of overlap. Agents do well across a wide range of innovation probabilities, especially when copying is between low and intermediate (~0.25-0.30). Copying becomes more advantageous as the probability of innovation increases, likely because there are more novel routes available to learners. There are also some important discrepancies between best outcomes in terms of minimum and median times. When innovation is ubiquitous (probability of 1) the minimum times get better while the median times get worse, relative to lower innovation probabilities (left and middle columns of Figure 4). This is probably because good solutions do not have a chance to spread without being modified. The copying probability at which minimum times and median times improve is also different. Average agents perform better with more copying (~0.25-0.30), whereas the best can do well at lower values (~0.1). This may be because the best agents are innovating novel solutions, while average agents rely on copying existing solutions.

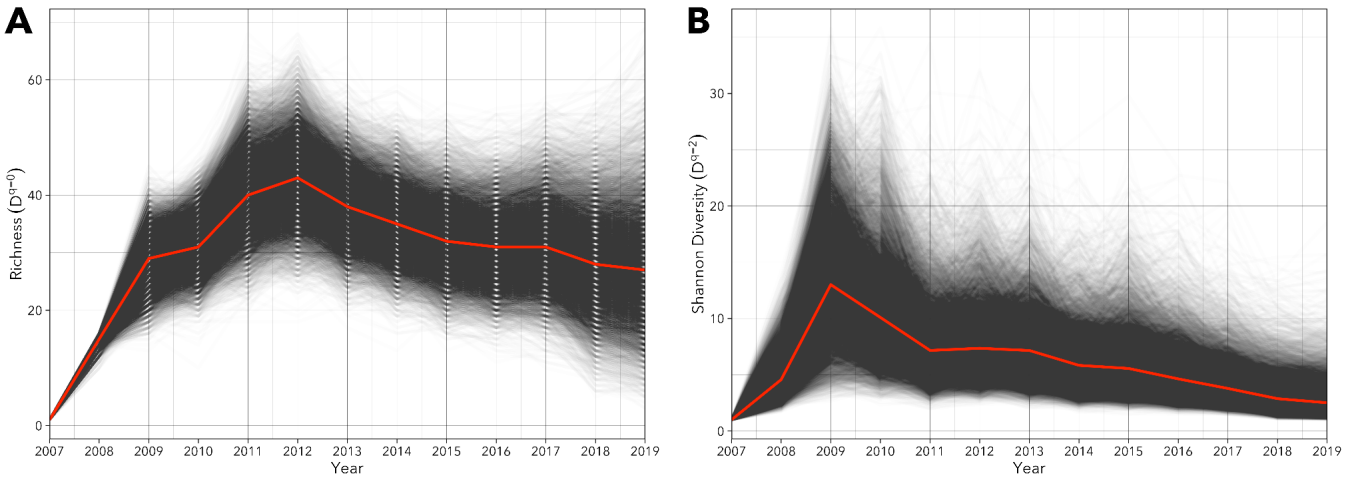


Figure 5. The richness (Hill number with q = 0; panel A) and Shannon diversity (Hill number with q = 1; panel B) of routes over time from 10,000 posterior simulations of the fitted model. Shannon diversity was used because it emphasizes rare items more than Simpson diversity (Hill number with q = 2), and many routes only appear once.

The fitted level of innovation and copying (black dot) appears to balance some of these trade-offs, even if the outcomes are suboptimal according to the model. The estimated level of copying is just above and squarely within the optimal zones for the best and average agents, respectively. On the other hand, the model estimates that much higher levels of innovation would lead to better outcomes for both the best and average agents. It is important to note, though, that there are many other constraints, not included in the model, that restrict the true solution space of the sport (e.g., acceleration, foot placement, height differences). This makes it difficult to conclude whether the observed departures from optimality reflect suboptimal strategies among agents or an underspecified model that does not accurately represent the physical constraints (49).

Interestingly, the zones with the lowest climbing times do not cleanly map onto the zones with the highest route diversity (right column of Figure 4). An intermediate level of innovation with no copying leads to a situation where route diversity is maximized (right column of Figure 4) but the solutions are extremely poor (left and center column of Figure 4). As copying increases, the diversity of routes goes down (right column of Figure 4) but the solutions improve (up to a point; left and center column of Figure 4). This could reflect population-level trade-offs between exploration and exploitation (50)—there is probably a "sweet spot" of route diversity that allows athletes to converge on reasonably good solutions. Figure 5 shows the richness and diversity of routes in the fitted model over time, and there appears to be an initial ~2-5 year increase in solutions followed by a ~7-10 year decay. This suggests that agents may explore a larger number of options before settling on, and iteratively improving upon, a smaller set of solutions. It is also worth noting that the model overestimates the number of routes in 2019, around ~24 (Figure 5A). The real estimated number of route sequences in 2019—extrapolated from the 2019 World Championship data using an unseen species model—is 15.02 (95% CI: [3.51-26.53], from 1000 bootstrap samples) (51).

The results of our analysis of manually-transcribed route sequences from the five IFSC World Championships between 2012-2019 can be seen in Tables 2 and 3. First, it appears that age, career length, height, and weight do not have a systematic effect on climbing times (Table 2). These same variables, alongside gender and a climber's rank in 2016, do not have an effect on whether climbers adopted the "Reza" in 2018 and/or 2019 (Table 4). Height and weight do, however, affect the use of some holds (Table 3, complete version in SI). Taller and heavier climbers are more likely to use #2—the optional higher starting hold—and less likely to use #14. Taller climbers are also less likely to use #5. It is important to note, though, that the usage of most holds (62.5%) is unaffected. Since these demographic factors only affect the usage of a minority of holds, do not affect climbing times or the adoption of the "Reza", and are not available for a large majority of athletes (outside of this subsample), we are confident in our decision to exclude them from the agent-based model.

Parameter	**Est.**	**Est. Error**	**2.5%**	**97.5%**
Age	-0.179	0.096	-0.368	0.011
Career	0.074	0.107	-0.136	0.283
Height (cm)	0.039	0.122	-0.180	0.294
Weight (kg)	-0.083	0.114	-0.309	0.123

Table 2. The effect of age, career length, height, and weight on climbing time, from a linear model with gender, name, and year as varying intercepts. 2.5% and 97.5% denote the bounds of the 95% credible interval, and intervals that do not overlap zero are marked with *.

	Age	**Career**	**Height**	**Weight**	**Gender (W)**
Hold 2	0.521	0.105	2.222*	1.457*	0.065
Hold 4	-0.670	1.127	0.352	-0.323	1.455
Hold 5	-0.155	-1.167	-1.911*	-1.329	0.651
Hold 9	-0.249	-0.113	-0.499	-0.458	0.410
Hold 10	0.035	0.044	0.649	0.642	-0.210
Hold 14	-0.119	0.298	-1.581*	-1.365*	0.530
Hold 15	0.267	0.354	0.460	0.538	-0.346
Hold 16	-0.100	0.047	-0.952	-1.321	0.304

Table 3. The effect of age, career length, height, weight, and gender on the probability of hold usage, from a multivariate logistic model with name and year as varying intercepts. This is an abbreviated table that only shows the effect of each predictor (columns) on holds (rows), with estimates whose 95% credible intervals do not overlap zero marked with *. The full table can be seen in Table S3.

Parameter	**Est.**	**Est. Error**	**2.5%**	**97.5%**
Age	0.611	0.695	-0.707	2.021
Career	1.009	0.796	-0.508	2.609
Height (cm)	-0.055	0.813	-1.707	1.490
Weight (kg)	-0.620	0.889	-2.340	1.135
Gender (W)	0.396	0.900	-1.372	2.159
Past Rank	0.073	0.648	-1.210	1.343

Table 4. The effect of age, career length, height, weight, gender, and rank in 2016 on the use of hold #4 in 2018 and/or 2019, from a logistic model. Negative effects denote the adoption of the "Reza" (skipping hold #4). 2.5% and 97.5% denote the bounds of the 95% credible interval, and intervals that do not overlap zero are marked with *.

Discussion

Our agent-based model, fit to over a decade of professional competition and 7,724 times, yields several interesting insights about the trade-offs between copying and innovation in speed climbing. First, the agents in our model are about 1.2 times as likely to copy an existing route than to innovate a new one, suggesting that the payoff for copying only slightly outweighs the payoff for innovation (Figure 2, but the 1.2 value comes from actual rather than attempted copying and innovation in the model). Copying allows climbers to rapidly adopt the best routes being used by other competitors, but some level of innovation is required to introduce new solutions. Second, innovation rates appear to decrease over time and with increasing population size, and slower agents are more likely to innovate than faster agents. The temporal dynamics of innovation are apparent in the timeline shown in Figure 5, where a rapid increase in the diversity of routes in the early years of competition—presumably from innovation—is followed by a slow decay in diversity as the agents converge towards a smaller set of more optimal routes. The effect of ability on innovation is also intuitive, as slower climbers who are not benefitting from existing routes can get a huge payoff from new solutions. Finally, a parameter search of alternative combinations of innovation and copying suggests that it would be beneficial for agents to innovate even more to improve their climbing times, even at the expense of a slight reduction in copying (Figure 4). This final result should be interpreted with caution, though, as the real solution space is likely quite different than the one in our model due to the myriad other physical constraints that we were unable to include (see below). Despite these limitations, the fitted version of our model is able to reproduce key features of the real data: the distribution of climbing times, the number of holds skipped, and the most common route sequences (Figure 3).

The analysis of manually-transcribed route sequences from 2012-2019 IFSC World Championships yields some additional insights. Height and weight only affected the use of a minority of skipped holds (Table 3). Otherwise, demographic factors—including height, weight, age, gender, and career length—did not influence climbing times (Table 2) or the adoption of the "Reza" (skipping hold #4; Table 4). These findings validate our decision to exclude these factors from our agent-based model, and focus instead on simpler social learning strategies.

Social learning is thought to be one of the key drivers of human cultural complexity (7). More specifically, combining social learning with relatively lower levels of innovation is advantageous because it allows improvements to rapidly spread (7,8). The utility of this strategy has been investigated in empirical work (10,11) is now used to explain the complexity of a wide range of cultural behaviors, such as food processing (52). However, the cumulative improvement of solutions to new problems is often framed as a cooperative goal (53), with relatively less attention paid to competition (9,54). Simulations of the learning vs. innovation trade-off show that competition dramatically increases the incentive to find unexploited regions of the solution space (9). In speed climbing, and perhaps in sports more generally, competition is the primary driver of improvement. This may explain why the estimated innovation rate in our model is relatively high, which is consistent with other competitive contexts where social information has only a slight advantage over personal information (10–12), or is even disadvantaged (13). Interestingly, agents in our model appear to be over-relying on copying and under-relying on innovation (Figure 4)—a result that runs counter to other work showing that humans under-utilize social information across a wide range of contexts (55,56).

Population size is known to predict various dimensions of cultural complexity, such as technological complexity (25) and linguistic complexity (57). In our fitted model, population size negatively predicts innovation rate, presumably because agents do not need to rely on innovation as much when there are already plenty of existing solutions to choose from. Interestingly, though, the agents in our model do not optimize for route diversity (right panel Figure 4). Instead, they converge towards a plateau level of diversity after an initial period of exploration (Figure 5), as also occurs in programming contests (6). This may reflect a general tendency towards *transient diversity* in collective problem-solving in humans—populations are more likely to discover good solutions when there is a baseline level of diversity and turnover, especially at the beginning of the process (53). However, too much diversity can cause populations to be slower in adopting the most efficient solutions (53). Our model shows that diversity is maximized (~80 routes) when innovation occurs half of the time and copying is rare, but this leads to extremely poor climbing times (Figure 4). The "sweet spot" for most climbers appears to be an innovation probability of ~0.7 and a copying probability of ~0.3, which leads to slightly above-average diversity (~35 routes; Figure 4).

Our fitted agent-based model suggests that climbers may adjust their innovation strategies based on their ability—slower agents are more likely to innovate than faster agents. This finding is consistent with decision-making studies showing that humans and other animals are more likely to search for new solutions (i.e., switch from an exploit to an explore strategy) when they are underperforming on a task, and vice versa (18,20). Simulation work in cultural evolution also suggests that a "critical social learning" strategy, where agents switch from social to asocial learning (i.e., from copying to innovating) when they underperform, outcompetes other strategies across a wide range of conditions (19,21). Interestingly, agents in our model do not shift their learning strategies in the same way that they shift in innovation. Slower agents are not more or less likely to copy than faster agents. This contrasts with findings from other competitive contexts. The best contestants in programming competitions are more likely to copy others (15), and the best players of the ancient board game Go are less likely to copy others (12). However, there is huge variation in strategies across competitors, and in some cases, such as tactics among football coaches, the relative use of social information is

decoupled from individual performance (13). At first glance, it seems strange that agents would only adjust innovation based on their performance. Innovation and copying are not zero-sum in our model, though—agents are able to both copy and innovate within a single timestep—and experimental work in humans suggests that one's propensity to innovate does not predict one's propensity to copy (58).

Several limitations of this study should be highlighted. First, we only have access to competition results, whereas climbers primarily try out new route sequences during training. Competitions occur frequently enough (every ~2-3 months) that new sequences innovated during training should be observed relatively quickly. However, each instance of innovation and copying in our model should be considered to be the outcome of an entire year of training and competition. Second, we include both women and men in our model to avoid confounding between-gender copying with within-gender innovation. Even though women and men are allowed to vary in their rate of athletic improvement, there are other factors, like height, that affect climbing strategy (1) and are not included in our model. Finally, we make several simplifying assumptions to reduce the dimensionality and computational cost of our analysis. Most notably, our model ignores foot positioning and velocity, both of which heavily constrain what is physically possible in speed climbing (2,4,5). This means that our measures of route diversity may be systematically overestimated (Figures 4 and 5). Despite these simplifying assumptions, we think that the model provides important insights into innovation and copying processes in speed climbing, and in cultural evolution more generally, as it is able to reproduce key features of contemporary route sequences (Figure 3). Emerging technologies such as force sensors (37) and motion tracking (4,5,34–36) may make it possible to include more realistic physical constraints in future studies.

In summary, this study provides new insights into the processes that drive cumulative improvement in competitive speed climbing. Our agent-based model, fit to over a decade of climbing times, suggests that innovation and copying are used roughly equally by climbers, with copying having only a slight advantage. Simulated histories indicate that there was likely a rapid expansion in the number of routes in the first few years of the sport, followed by a slow convergence towards a smaller set of more efficient solutions. Slower agents in our model are more likely to innovate, likely in pursuit of strategies that give them a competitive advantage. Population size negatively predicts innovation, presumably because innovation is not as useful when there are plenty of existing solutions to choose from. The model also suggests that climbers may not be using innovation and copying optimally—continued improvement may require more emphasis on innovation and less reliance on copying. Despite several major limitations, including the exclusion of several key physical constraints like foot position and velocity, our model is able to approximate key features of speed climbing including the distribution of climbing times in each year, the average number of holds skipped, and common route variations like the "Reza".

Data & Code Availability

All data and code can be found in the GitHub repository: https://github.com/masonyoungblood/SpeedClimbingABM

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Supplementary information

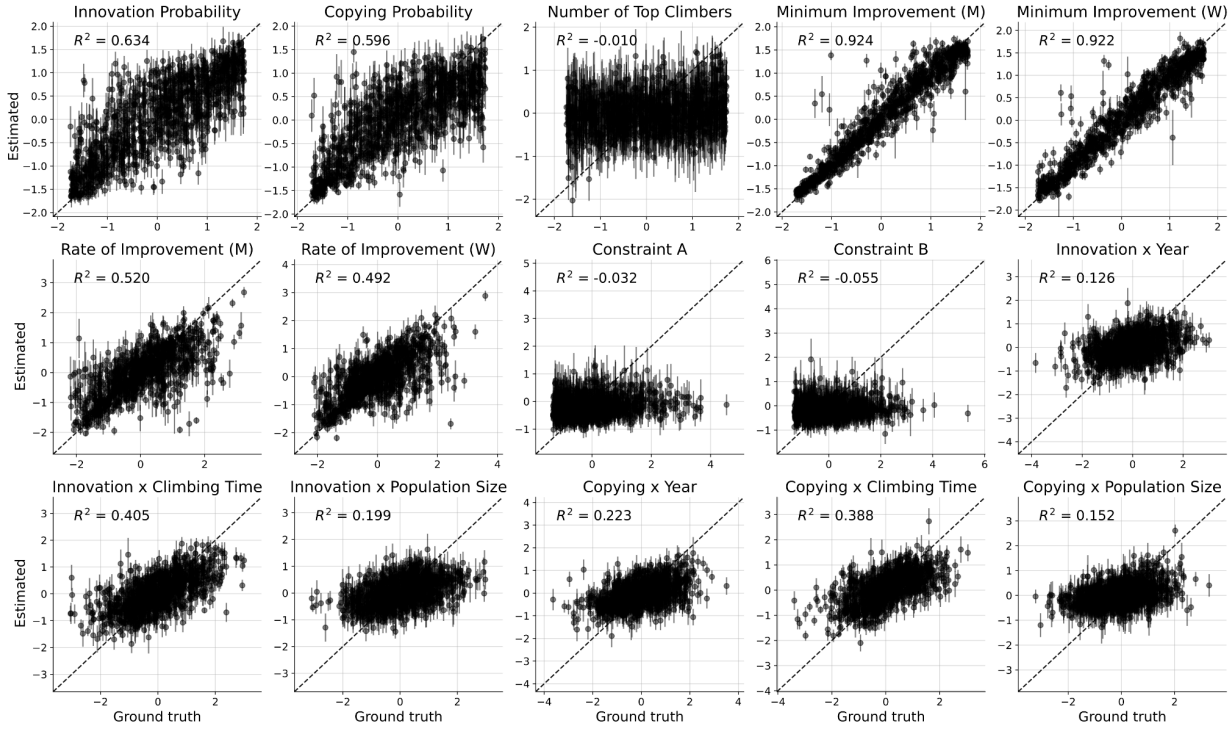


Figure S1. Recovery plots showing BayesFlow's ability to recover known values of each parameter in the agent-based model. The x-axis corresponds to z-scored known values of each parameter, and y-axis corresponds to z-scored estimated values of each parameter. Each point estimate includes error bars—the median absolute deviation of 100 posterior samples. The dotted line marks the expectation assuming perfect prediction ($R^2 = 1$). This figure shows the results of BayesFlow trained on 10,000 iterations of the agent-based model (90% training set and 10% testing set) when the two constraints on innovation were included (second row, third and fourth columns).

We experimented with incorporating two plausible biases in innovation: (1) a preference for skipping holds that lead to shorter gaps in the route, and (2) a preference for skipping holds that make the path more direct. The probability of skipping each hold *P(h)* was computed as follows:

$$P(h) = \frac{rank(D_h)^{-c_D} \cdot rank(R_h)^{-c_R}}{\sum_{h=1}^{20} rank(D_h)^{-c_D} \cdot rank(R_h)^{-c_R}}$$

D_h is the Euclidean distance between the adjacent holds (the holds used before and after hold h) and R_h is the ratio of the original to the new cumulative path length. D_h and R_h are both ranked (ascending) and raised to parameters c_D and c_R that control their relative impact of the probability of holds being skipped. Both of these parameters can take positive and negative values, corresponding to preference for or against skipping holds that lead to shorter gaps or more direct paths in the route. Neither c_D nor c_R could be recovered by simulation-based inference (see Simulation-Based Inference in Methods and Figure S1), so biased innovation was removed from the model—all holds are equally likely to be skipped.

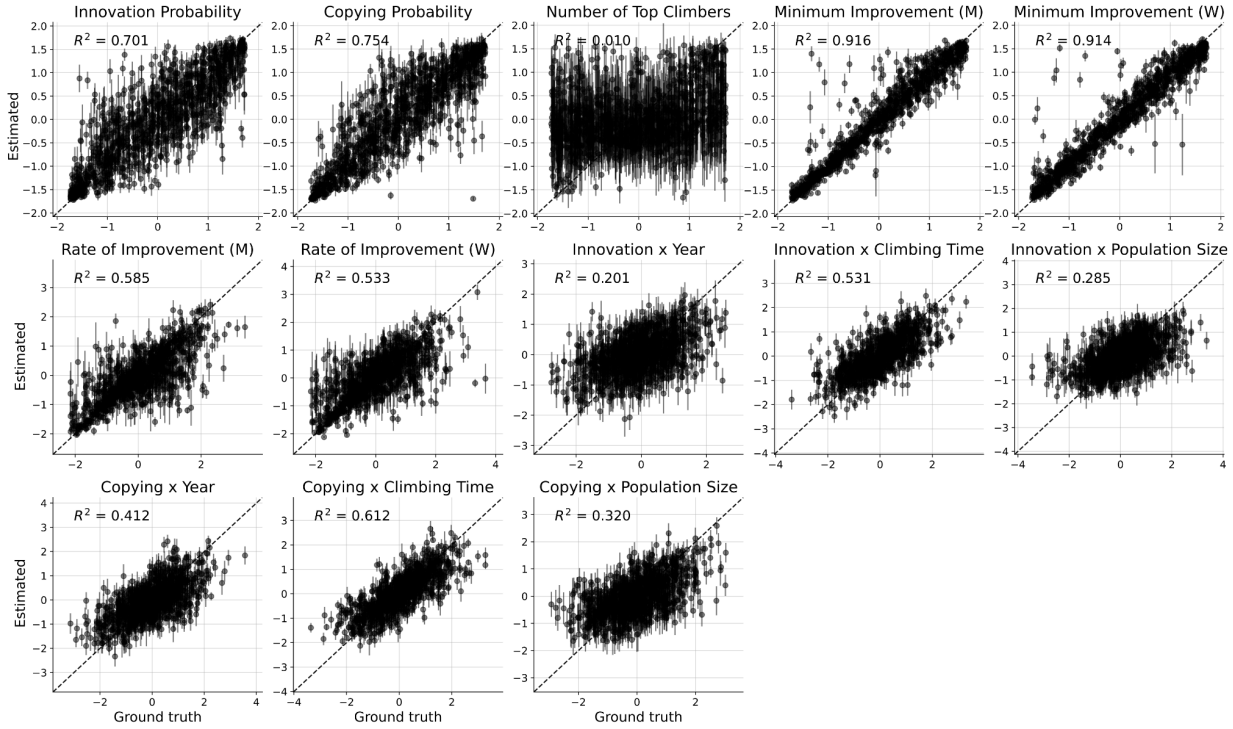


Figure S2. Recovery plots showing BayesFlow's ability to recover known values of each parameter in the agent-based model. The x-axis corresponds to z-scored known values of each parameter, and y-axis corresponds to z-scored estimated values of each parameter. Each point estimate includes error bars—the median absolute deviation of 100 posterior samples. The dotted line marks the expectation assuming perfect prediction ($R^2 = 1$). This figure shows the performance of the final inference model—BayesFlow trained on 100,000 iterations of the agent-based model (99% training set and 1% testing set) when the two constraints on innovation were excluded (see Figure S1).

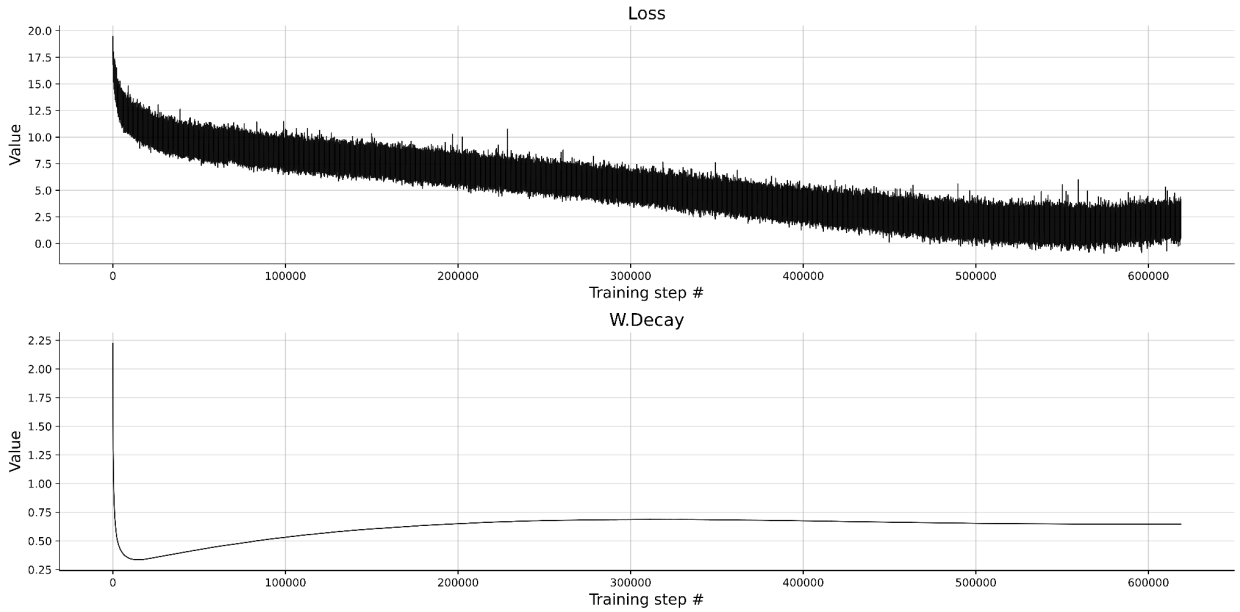


Figure S3. The training loss (top) and weight decay (bottom) of the final inference model—BayesFlow trained on 100,000 iterations of the agent-based model (99% training set and 1% testing set).

Age	Career	Height	Weight
1.000	1.000	1.000	1.000
1.037	1.012	1.033	1.024
1.003	1.000	1.003	1.005
1.003	1.001	1.000	1.000
1.002	1.003	1.001	1.002
1.002	1.000	1.000	1.001
1.004	1.001	1.000	1.001
1.001	1.001	1.001	1.000
1.002	1.006	1.003	1.000
1.007	1.008	1.000	1.001

Table S1. The R-hat diagnostic values from the linear model predicting climbing time, for each parameter (columns) and each of the 10 imputed datasets (rows). Values near 1 indicate that the MCMC chains have converged.

Age	Career	Height	Weight	Gender	Past Rank
1.003	1.001	1.000	1.001	1.000	1.001
1.000	1.000	1.000	1.002	1.000	1.001
1.001	1.001	1.000	1.000	1.000	1.000
1.000	1.000	1.000	1.001	1.000	1.002
1.000	1.000	1.002	1.001	1.000	1.000
1.000	1.000	1.000	1.000	1.001	1.000
1.001	1.000	1.000	1.000	1.000	1.000
1.000	1.000	1.000	1.001	1.001	1.000
1.000	1.002	1.000	1.000	1.000	1.000
1.000	1.000	1.000	1.000	1.002	1.001

Table S2. The R-hat diagnostic values from the logistic model predicting the adoption of the "Reza" (skipping hold #4), for each parameter (columns) and each of the 10 imputed datasets (rows). Values near 1 indicate that the MCMC chains have converged.

Hold	Parameter	Estimate	2.5%	97.5%		Avg. R-hat
2	Age	0.401	-0.859	1.721		1.000
	Career	0.212	-1.107	1.586		1.001
	Height	2.100	0.695	3.478	*	1.001
	Weight	1.201	-0.148	2.684		1.000
	Gender (W)	-0.040	-1.641	1.487		1.000
4	Age	-0.494	-1.737	0.852		1.000
	Career	1.094	-0.164	2.358		1.000
	Height	0.082	-1.131	1.202		1.000
	Weight	-0.281	-1.674	0.885		1.000
	Gender (W)	1.260	-0.337	2.809		1.001
5	Age	-0.036	-1.444	1.334		1.000
	Career	-1.277	-2.677	0.084		1.001
	Height	-1.915	-3.213	-0.755	*	1.000
	Weight	-1.359	-2.857	0.046		1.000
	Gender (W)	0.515	-1.262	2.309		1.000
9	Age	-0.434	-2.063	1.201		1.000
	Career	-0.056	-1.727	1.586		1.000
	Height	-0.461	-2.154	1.163		1.001
	Weight	-0.555	-2.169	1.011		1.001
	Gender (W)	0.380	-1.453	2.210		1.001
10	Age	0.170	-1.406	1.811		1.000
	Career	0.020	-1.586	1.657		1.000
	Height	0.635	-0.978	2.339		1.000
	Weight	0.614	-0.928	2.245		1.000
	Gender (W)	-0.187	-2.032	1.634		1.000
14	Age	0.021	-1.348	1.403		1.000
	Career	0.200	-1.140	1.483		1.000
	Height	-1.633	-2.910	-0.471	*	1.001
	Weight	-1.386	-2.809	-0.016	*	1.001
	Gender (W)	0.361	-1.391	2.149		1.000
15	Age	0.137	-1.057	1.366		1.000
	Career	0.462	-0.726	1.689		1.000
	Height	0.570	-0.480	1.683		1.001
	Weight	0.409	-0.754	1.615		1.000
	Gender (W)	-0.316	-1.839	1.205		1.000
16	Age	-0.020	-1.471	1.462		1.000
	Career	-0.007	-1.443	1.398		1.000
	Height	-0.978	-2.286	0.226		1.001
	Weight	-1.245	-2.736	0.200		1.001
	Gender (W)	0.256	-1.537	2.083		1.000

Table S3. The effect of age, career length, height, weight, and gender on the probability of hold usage, from a multivariate logistic model with name and year as varying intercepts. Positive and negative values for gender correspond to women and men, respectively. 2.5% and 97.5% denote the bounds of the 95% credible interval, and intervals that do not overlap zero are marked with *. The R-hat diagnostic values are averaged across the 10 imputed datasets. R-hat values near 1 indicate that the MCMC chains have converged.